Original scientific paper

European Journal of Preventive Cardiology

European Journal of Preventive Cardiology
0(00) 1–10

DOI: 10.1177/2047487316634883
ejpc.sagepub.com
SAGE

Validation of Fitbit-Flex as a measure of free-living physical activity in a community-based phase III cardiac rehabilitation population

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Abstract

Background: Accurate physical activity monitoring is important for cardiac patients. Novel activity monitoring devices may enable precise measurement of physical activity. This study aimed to validate Fitbit-Flex against Actigraph accelerometer for monitoring physical activity.

Design: A validation study with a comparative design.

Methods: Cardiac patients and family members participating in community-based exercise programs wore Fitbit-Flex and Actigraph simultaneously over four days to monitor daily step counts and minutes of moderate to vigorous physical activity (MVPA).

Results: Participants ($N = 48$) comprised 52.1% males, with a mean age of 65.6 ± 6.9 years and 58.9% had a cardiac diagnosis. Fitbit-Flex and Actigraph were significantly correlated in males, females, total participants and cardiac patients for step counts ($r = .96$; $r = .95$; $r = .95$; $r = .95$), though less so for MVPA ($r = .81$; $r = .65$, $r = .74$; $r = .71$). As step counts increased the differences between Fitbit-Flex and Actigraph also increased. Fitbit-Flex over-estimated step counts in females (556 steps/day), males (1462 steps/day) and total participants (1038 steps/day) as well as for minutes of MVPA in females (4 min/day), males (15 min/day) and total participants (10 min/day). Fitbit-Flex had high sensitivity and specificity in classifying participants who achieved the recommended physical activity guidelines.

Conclusion: Fitbit-Flex is accurate in assessing attainment of physical activity guideline recommendations and is useful for monitoring physical activity in cardiac patients. The device does, however, slightly over-estimate step counts and MVPA.

Keywords
Coronary heart disease, physical activity, validity, Actigraph, Fitbit, accelerometer, activity monitor, cardiac rehabilitation

Received 29 October 2015; accepted 4 February 2016

Introduction

Physical activity is a major component in a cardiac rehabilitation programme and is integral to recovery from coronary heart disease (CHD). This is primarily because physical activity appears to mitigate the progression of CHD.[1] A substantial body of evidence shows that an increase in physical activity levels results in significant improvements in many well-known CHD risk factors including hypertension,[2] dyslipidaemias,[3] obesity,[4] insulin resistance[4] and psychological health.[5] However, these health benefits are only achieved when patients engage in the recommended total weekly amount of physical activity of at least moderate-intensity level for 150 min/week or more.[6] One of the key aims of cardiac rehabilitation is to encourage patients to achieve these recommended levels of physical activity. Cardiac rehabilitation also has medical, psychological, social and behavioural benefits and can increase functional capacity.[1,7] Interestingly, the benefits of cardiac

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rehabilitation may include partners of patients with CHD as cardiac risk factors are often shared.[8,9] Therefore, promotion of physical activity in the family members of patients with CHD attending cardiac rehabilitation is also worthwhile.

Physical activity involves a complex set of behaviours which may not easily be measured. The most common measures in cardiac rehabilitation are self-reported questionnaires that are prone to reporting errors and social desirability bias.[10] A systematic review of self-report physical activity measures for cardiac patients reveals most of the instruments have great variability, poor validity and reliability, and are more suited to epidemiologic studies rather than assessing intervention outcomes where responsiveness to intervention is vital.[10] Accelerometry-based activity monitors – devices that detect acceleration in one (uni), two (bi) or three (triaxial) directions, and which measure the quantity and intensity of movements – have become standard objective methods for assessing physical activity.[11,12] Triaxial accelerometers such as the Actigraph are regarded as the gold standard for measurement of physical activity.[13,14] However, these devices are expensive, uncomfortable to wear and are not able to provide feedback to the wearer.

Use of wearable physical activity monitors offers significant promise to researchers and clinicians working to promote or measurer physical activity.[15,16] Activity tracking devices have the potential to effect physical activity behaviour change, make a direct and real-time impact on self-management of physical activity and offer clinicians real world assessments of their patients' daily activity patterns.[15] New physical activity trackers regularly enter the market, with advances in technology contributing to dramatic improvements, including inbuilt accelerometers for smartphone-based devices and wearable physical activity devices such as Fitbits. Despite their growing popularity, however, there is limited evidence in research and clinical settings on the validity and reliability of these devices. Notably, different validity criteria exist in the literature for the measurement of physical activity between the tested instrument and the criterion instrument based on the purpose of the measurement. For instance, the validity criteria between the two instruments for clinically acceptable agreement is per cent error of less than 20%,[17] whereas the validity criteria to achieve acceptable agreement for research purposes is per cent error within 3%.[18,19]

Fitbit-Flex is one of the most popular commercial wearable devices currently available to measure physical activity. It is inexpensive, worn on the wrist, and accessible to the public. However, research on the accuracy of Fitbit devices for measuring physical activity is limited.[15,20-22] Ferguson et al.[15] validated Fitbit-Zip and Fitbit-One against Actigraph GTX3+ accelerometer in healthy participants who wore both monitors simultaneously for two days. The results showed the Fitbit devices were valid for measuring step counts. To our knowledge, no study has previously examined the validity of Fitbit in estimating step counts and active minutes in cardiac rehabilitation participants.

It is important to assess physical activity in cardiac rehabilitation participants because compared with the general population they are more likely to have lower levels of activity,[23,24] are often of older age and have conditions that cause symptoms such as chest discomfort, dizziness, shortness of breath and leg or arm discomfort.[25] Fitbit devices may be ideal to use in this population to help track physical activity and to motivate sustained changes in moderate-intensity physical activity. Therefore, the primary aim of this study is to validate Fitbit-Flex against a gold standard instrument (Actigraph GT3X triaxial accelerometer) for comparing average daily step counts and minutes per day of moderate to vigorous physical activity (MVPA) among cardiac patients and eligible family members who are enrolled in community-based phase III cardiac rehabilitation. Within this aim, this study assesses the sensitivity, specificity, positive predictive value (PPV) and area under curve (AUC) of Fitbit-Flex for predicting attainment of recommended physical activity levels. The secondary aims of this study are to validate Fitbit-Flex for measuring other active minutes of physical activity and to examine the independent predictors of mean differences for step counts and MVPA.

Methods

Study design

A validation study with a comparative design was utilised to assess a wearable physical activity measurement device (Fitbit-Flex) versus the gold-standard measurement triaxial accelerometer (Actigraph).

Subjects and setting

Participants diagnosed with CHD and their family members were recruited from two community exercise groups connected to North Sydney Local Health District, Sydney, Australia. Both groups comprised previous cardiac rehabilitation patients, family members and friends. One group is a community-based phase III cardiac rehabilitation programme and engages in a structured programme of five classes per week of group-based traditional exercise and aqua classes led by an exercise physiologist or a cardiac rehabilitation clinical nurse specialist. The classes involved resistance training, balance and coordination training, functional movement exercises, and activities to improve aerobic endurance. The other group is an independent volunteer-walking group and engages in a fortnightly programme of social walks of 1–2 h duration. Participants set their own pace and choice of duration.

Exercise group members were eligible for the study if: 1) they or their family member had a diagnosis of CHD and had completed a phase II cardiac rehabilitation programme; 2) engaged in regular physical activity equivalent to 30 minutes per day; 3) had sufficient English to understand the consent process and other study requirements; 4) were willing to participate for the entire four-day study period (assessed at initial interview); and 5) were able to apply and wear both the Fitbit-Flex and Actigraph devices simultaneously. Group members were excluded if they had major co-morbidities (e.g. unstable angina, uncontrolled hypertension, uncontrolled arrhythmia, any neurological cognitive disorder) that would limit their ability to participate in regular physical activity during the study period or affect their ability to apply the devices. Ethics approval was obtained from the university research ethics committee. The study was conducted from December 2014 to February 2015.

Primary outcomes were the mean differences as measured by Fitbit-Flex and Actigraph in the daily averages of the number of steps and the number of minutes spent participating in MVPA. Prior studies of the correlation between activity monitor output and observed step counts reported a range from 0.5 to 1.0.[26] A priori sample size of 23 participants was calculated based on the most conservative findings (correlation of 0.5), alpha level = 0.05, and a power of 0.80. However, our results from the Healthy Eating Exercise Lifestyle Program indicated that female participants reported statistically significantly lower exercise duration than males.[27] Therefore, results must be analysed separately by gender so a minimum of 46 participants (23 males and 23 females) was required.

Data collection and measures

Data were collected at baseline and at completion of the four days of physical activity measurement.

Demographic and clinical data

Participant demographic and clinical data were self-reported and collected using a data collection form created for this study. Demographic details included marital status, gender and level of education. Clinical data included medical history of co-morbidities and CHD.

Anthropometric measurements

Waist circumference, height and weight were self-reported where the participant could report a recent, accurate measure. Where they could not, or were uncertain of these measures, they were undertaken by the researchers. For the measures, all participants wore light exercise clothing and were asked to remove their shoes.

Physical activity

Physical activity for daily step counts and moderate vigorous minutes were measured using Fitbit-Flex and Actigraph accelerometer.

Fitbit-Flex

Fitbit-Flex (Fitbit Inc., San Francisco, CA, USA) is a small (140–176 mm) and light (8 g) accelerometer. Fitbit devices were reported to be valid instruments to measure daily caloric expenditure among older adults[28] and step counts among healthy adults.[21] The devices are reported by the manufacturers to sense motion in three dimensions and to convert these readings into activity information.[28] The tracker measures daily steps, distance, calories burned and overall activity level (intensity and duration). Fitbits synchronise wirelessly to either an installed mobile phone application (via Bluetooth) or an installed computer application (via USB dongle). Synchronising Fitbits enables data to be sent to a secured website, Fitbit.com, where it is recorded.

Actigraph GT3X triaxial accelerometer

Actigraph GT3X triaxial accelerometer (model 7164, Manufacturing Technology, Inc., FL, USA) is small (4.6 cm × 3.3 cm × 1.5 cm) and light (19 g). It was worn around the waist and was reported to be a reliable and valid objective measure of physical activity among cardiac patients after a cardiac rehabilitation programme.[29] The device records raw acceleration information as an activity count that may be used to estimate the intensity and duration of body movement.[6] The Actigraph accelerometer, in conjunction with the ActiLife software application, produces physical activity measurements including energy expenditure, steps taken and physical activity intensity.[6] The devices were initialised for participants using the manufacturer's software, ActiLife Version 6. The participants' actual characteristics (height and weight) were entered into the software programme and the data were then downloaded.

Procedure

Potential participants were informed of the study while attending community exercise classes at a local sports club and the community walking group at a nearby public reserve. People who were then screened as eligible and consented to join the study undertook the baseline survey. Height, weight and waist circumference were measured or self-reported. Each participant received the Fitbit-Flex and Actigraph devices pre-initialised to collect data using 1-min epochs. Instructions on monitor placement and wearing time for each device were provided. To estimate habitual physical activity participants were instructed not to change their exercise routine. Participants were then asked to simultaneously wear Fitbit-Flex and Actigraph for four consecutive days (two weekend days and two week days) from the time they awoke in the morning until the time they went to bed at night, excepting water-based activities such as swimming or showering. Data from the accelerometers were downloaded at completion.

Of the 48 participants, 23 were females and 25 males, 28 had a cardiac diagnosis and 20 were family members. In terms of recruitment, 38 participants were recruited from a structured community-based phase III cardiac rehabilitation programme and 10 were recruited from a community walking group. Forty-four (91.7%) had complete data for the whole four-day study period. One participant lost the Fitbit-Flex device and was excluded from the study.

Data analysis

Data analyses were performed using the Statistical Package for the Social Sciences (SPSS) Version 22. Means, standard deviations, frequencies and percentages were used to describe the socio-demographic, clinical and physical activity characteristics of the sample.

Physical activity classification of sedentary, low, moderate, vigorous or MVPA were based on classification criteria derived from Actigraph accelerometer and Fitbit-Flex data. Actigraph data were categorised into count per minute thresholds to describe daily activity: sedentary (0 to 100); light intensity (101 to 2020); moderate intensity (2021 to 5999); vigorous intensity (≥ 6000); and MVPA (≥ 2021).[30] However, Fitbit-Flex used metabolic equivalents (METs) to calculate daily activity: sedentary (a MET value of 1 represents resting); moderate intensity (for activities at or above 3 METs); and vigorous intensity (for activities at or above 6 METs).[31] The Fitbit website does not provide clear explanation of how these thresholds are defined or calculated.

Pearson correlation coefficients were used to assess the strength of the relationship between Fitbit-Flex and Actigraph in measuring step counts and active minutes for males and females separately. Bland–Altman plots were used to illustrate levels of agreement between Fitbit-Flex and Actigraph in recording step counts. Paired *t*-test was used to detect differences between Fitbit-Flex and Actigraph in measuring absolute differences in average daily step counts and daily minutes of MVPA over four days. Sensitivity, specificity, PPV and AUC were calculated for physical activity cut-off points of 7000 and 10,000 steps/day,[32] and 150 min/week[33] to identify those who met physical activity guideline recommendations. Linear regression analysis was also conducted to determine any independent predictors of the mean difference in total step counts and total minutes of MVPA over four days.

Results

Sample characteristics

Participants comprised 52.1% males, with a mean age of 65.6 ± 6.9 years, a mean waist circumference of 98.5 ± 6.4 cm and a mean body mass index (BMI) of $27.9 \pm 2.4\,kg/m^2$ (Table 1). More than half (58.9%) the participants had a cardiac diagnosis. Bone or joint problems (39.6%) were the most common co-morbidity impacting physical activity participation.

Correlations between step counts and active minutes for Fitbit-Flex and Actigraph

Regardless of gender and cardiac diagnosis, Fitbit-Flex is significantly correlated with Actigraph in measuring step counts and active minutes of light activity, moderate activity and MVPA (Table 2).

Physical activity-related characteristics: step counts and active minutes

The mean difference in participants' step counts per day and minutes of MVPA per day as derived from Fitbit-Flex and Actigraph indicated no statistically significant difference for effects of day across the four-day monitoring period. Fitbit-Flex and Actigraph varied in the step counts recorded for each day, but on all occasions Fitbit-Flex counted higher steps than Actigraph. Of note, Fitbit-Flex recorded significantly more steps than Actigraph in females by 7% (or 556 steps/day, $p = 0.01$; median = 459; interquartile range = 140–1000), males by 18% (or 1462 steps/day, $p = 0.01$; median = 958; interquartile range = 661–2251) and total participants by 13% (or 1038 steps/day, $p = 0.01$ median = 743; interquartile range = 419–1680).

Similarly, Fitbit-Flex estimated higher MVPA minutes compared with Actigraph in females by 4% (or 4 min/day, $p = 0.94$; median = 18; interquartile range = 7–32), males by 15% (or 15 min/day, $p = 0.01$; median = 16; interquartile range = 8–39) and total participants by 10% (or 10 min/day, $p = 0.01$; median = 17; interquartile range = 8–36). Generally, Bland–Altman

Table 1. Demographic and clinical characteristics ($N = 48$).

Characteristic	Total ($N = 48$)		Female ($n = 23$)		Male ($n = 25$)	
	Mean	SD	Mean	SD	Mean	SD
Age	65.6	6.9	62.0	5.7	68.8	6.4
Age range	(52–85)		(52–73)		(60–85)	
Body mass index (kg/m^2)	27.9	2.4	27.9	2.5	27.8	2.6
Waist (cm)	98.5	6.5	95.6	3.9	101.2	7.2
	Number	%	Number	%	Number	%
Cardiac diagnosis	28	58.9	8	34.8	20	80
Family members	20	41.7	15	65.25	5	20
Respiratory problems	5	10.4	1	4.2	4	16
Bone or joint problems	19	39.6	9	39.1	10	40
Married/partner	32	66.7	15	65.2	17	68
Bachelor degree or above	30	62.5	14	60.9	16	64
Caucasian	31	64.6	15	65.2	16	64

Table 2. Correlations between step counts and active minutes on Fitbit-Flex and Actigraph.

		Actigraph					
		Gender	Steps	Light	Moderate	Vigorous	MVPA
Fitbit-Flex	Steps (steps/4 days)	Males	.964[a]	.204	.793[a]	.608[a]	.860[a]
		Females	.945[a]	.294	.666[a]	.399	.666[a]
		Total	.948[a]	.194	.718[a]	.558[a]	.769[a]
		Cardiac only	.947[a]	–.049	.654[a]	.617[a]	.728[a]
	Light (minutes/4 days)	Males	.396	.752[a]	.659[a]	–.082	.587[a]
		Females	.284	.760[a]	.330	.155	.326
		Total	.330[b]	.762[a]	.526[a]	–.077	.476[a]
		Cardiac only	.205	.722[a]	.423[b]	–.065	.372
	Moderate (min/4 days)	Males	.638[a]	.503[b]	.754[a]	.151	.725[a]
		Females	.734[a]	.616[a]	.715[a]	.320	.706[a]
		Total	.660[a]	.489[a]	.718[a]	.180	.702[a]
		Cardiac only	.734[a]	.312	.670[a]	.289	.673[a]
	Vigorous (min/4 days)	Males	.458[b]	–.282	.375	.032	.351
		Females	.663[a]	–.057	.280	.199	.283
		Total	.537[a]	–.183	.326[b]	.073	.317[b]
		Cardiac only	.579[a]	–.365	.408[b]	.036	.380[b]
	MVPA (min/4 days)	Males	.768[a]	.376	.849[a]	.154	.813[a]
		Females	.834[a]	.442[b]	.658[a]	.325	.652[a]
		Total	.789[a]	.365[b]	.761[a]	.188	.743[a]
		Cardiac only	.839[a]	.116	.718[a]	.255	710[a]

[a]Correlation is significant at the 0.01 level (two-tailed).
[b]Correlation is significant at the 0.05 level (two-tailed).
MVPA: moderate to vigorous physical activity

plots for step counts demonstrated moderate agreement in females, males, total participants (Figure 1) and cardiac participants. Bland–Altman plots also showed that as step counts increased the difference between the Fitbit-Flex steps and the Actigraph steps also increased, but this is not likely to misclassify inactive people. The limits of agreement also tend to be wider, demonstrating greater variation in the recordings.

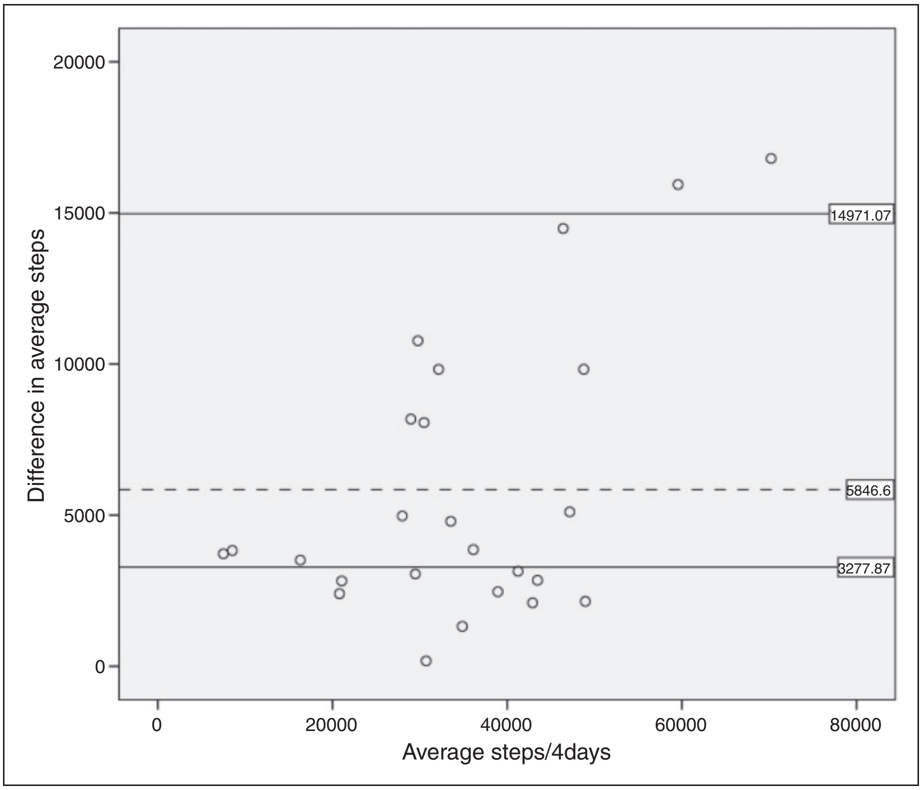

Figure 1. Bland–Altman plot (difference plot) of total steps measured by Fitbit-Flex and Actigraph (total participants). The dashed horizontal line corresponds to the mean difference (5846.6 steps/4 days) between Fitbit and Actigraph, while the solid horizontal lines correspond to the limits of agreement (3227.87 to 14971.07 steps/4 days) ($N = 47$).

Table 3. Sensitivity, specificity, positive predictive value of Fitbit-Flex for meeting physical activity guidelines compared with Actigraph.

Activity classifications	Sensitivity (95% CI)	Specificity (95% CI)	PPV (95% CI)	AUC (95% CI)
≥10,000 steps/day ($n = 11/17$)	1.00 (71.51; 100)	0.83 (67.91; 93.63)	0.65 (38.33; 85.97)	0.99 (95.00; 1.00)
≥ 7000 steps/day ($n = 26/35$)	1.00 (86.77; 100)	0.57 (34.02; 78.18)	0.74 (56.74; 87.51)	0.96 (92.00; 1.00)
≥ 150 min MVPA/week ($n = 44/45$)	1.00 (91.96; 100)	0.67 (9.43; 99.16)	0.98 (88.23; 99.94)	0.99 (97.00; 1.00)

CI: confidence interval; PPV: positive predictive value; AUC: area under curve; MVPA: moderate to vigorous physical activity

Fitbit-Flex was highly sensitive (100% accuracy) for identifying participants who achieved physical activity guidelines of 7000 and 10,000 steps/day cut-off points, and 150 min/week cut-off points, but the specificity and PPV of Fitbit-Flex for these cut-off points varied (Table 3). Fitbit-Flex had very high accuracy (AUC = 99%), and high (83%) and moderate (67%) specificity for identifying participants who failed to achieve at least 10,000 steps/day and 150 min/week or more of MVPA, respectively. However, Fitbit-Flex had moderate (57%) specificity, but very high accuracy (AUC = 96%) for identifying participants who failed to achieve at least 7000 steps/day. Among those who met physical activity guideline recommendations as classified by Fitbit-Flex, the probability of truly attaining the 150-min cut-off points was high (98%), but the probability of truly attaining the 7000 and 10,000 steps/day cut-off points was moderate (74% and 65%, respectively).

Independent predictors of the mean difference of step counts and MVPA as measured by Fitbit-Flex and Actigraph

The independent predictors of the mean difference for step counts and MVPA minutes were tested using a multivariate linear regression analysis with forced entry of the variables: age, gender and BMI. The full model for step counts containing all predictors was not statistically significant (F test = 2.43, $p < 0.08$, $R^2 = 0.145$). Similarly, the full model for MVPA minutes containing all predictors was not statistically significant (F test = 1.63, $p < 1.69$, $R^2 = 0.102$), and therefore details of the results are not reported.

Discussion

Overall, Fitbit-Flex is strongly correlated with Actigraph for measuring step counts among all examined subgroups, but the device appears to progressively over-estimate the step count as the number of steps increases. Fitbit-Flex also correlated with Actigraph for measuring all active minutes except for minutes of vigorous activity. Fitbit-Flex was very good at identifying participants who met physical activity guideline recommendations. Age, gender and BMI did not prove to be an important predictor of the mean difference for step counts and minutes of MVPA.

To our knowledge, this is the first study to examine the validity of Fitbit-Flex for monitoring physical activity in cardiac patients. However, other studies of similar activity monitors (e.g. Fitbit-Zip and Fitbit-One) exist. Our findings showed that Fitbit-Flex is highly correlated with Actigraph in measuring step counts. These results are difficult to compare with previous findings due to the relative paucity of research with the same physical activity measures (i.e. Fitbit-Flex). However, our results are consistent with the results of previous studies which found Fitbit devices are valid for measuring step counts in healthy participants,[15,16,21,34-36] and perform better than Fuelband devices in populations diagnosed with brain and stroke injury.[37] The findings of our study also demonstrated Fitbit-Flex is superior to the Omron pedometer for measuring step counts among cardiac rehabilitation participants.[24] Our finding extends previous research by showing that Fitbit-Flex can precisely measure step counts in free-living conditions over longer durations as opposed to treadmill walking for 25 minutes,[34] and has high accuracy in identifying participants who met physical activity guidelines for 7000 and 10,000 steps/day cut-off points. This is an important finding as it demonstrated the applicability of Fitbit-Flex for cardiac rehabilitation participants in measuring step counts.

Fitbit-Flex correlated significantly with Actigraph in measuring active minutes where no vigorous activity is incorporated. This finding was consistent with previous research which showed correlations existed between Fitbit-Zip and Actigraph, and Fitbit-One and Actigraph, in measuring minutes of MVPA.[15] Compared with cardiac rehabilitation studies, however, our finding showed Fitbit-Flex had substantially higher validity than three self-reported physical activity measures (Total Activity Measure (TAM);[38] International Physical Activity Questionnaire (IPAQ);[39] and a physical activity diary[40]) in assessing active minutes of physical activity. It should be noted that our results showed no significant statistical relationship between Fitbit-Flex vigorous activities and Actigraph vigorous activities. This may indicate that a different algorithm was used in each device, or the placement of Fitbit on the wrist rather than the waist may increase the measurement error when assessing vigorous intensity physical activity. Therefore, future studies are warranted to investigate the placement and the definition or the operationalised algorithm of vigorous activities for Fitbit devices, given intensity cut-off points are not provided by the Fitbit device manufacturers.

Fitbit-Flex can accurately identify participants who achieved physical activity guidelines of 150 min/week cut-off points in our study. Fitbit-Flex performed substantially better than the self-reported measure (Health Survey for England) in classifying cardiac rehabilitation participants who engaged in light, moderate and vigorous physical activity.[23] Notably, our physical activity level data (9224 ± 3907) demonstrated that this sample was very active compared with commonly reported data in cardiac rehabilitation research which indicates cardiac rehabilitation patients often exercise less than recommended guidelines for daily physical activity levels.[24,23] Nonetheless, the findings in this study suggested the applicability of Fitbit-Flex for cardiac rehabilitation participants in measuring MVPA.

The over-estimation of step counts and minutes of MVPA by Fitbit-Flex compared with Actigraph may be due to the manufacturer definition of the intensity cut-off points or the placement of Fitbit-Flex on the wrist rather than the waist. In comparison with previous research, a systematic review of male and female adults showed that self-reported measures over-estimated physical activity levels compared with accelerometers with a mean difference of 44%.[41] However, cardiac rehabilitation studies showed mixed results: compared with the accelerometer, the TAM self-report measure over-estimated time spent in physical activity by approximately 240 min/week,[38] whereas the IPAQ self-report measure under-estimated time spent in physical activity by approximately 153 min/day.[39] Similarly, for free-living physical activity in healthy adults, Schneider et al.[18] found large mean errors between 13 models of pedometer in measuring step counts over 24 h. Some pedometers under-estimated step counts by 25%, whereas others over-estimated step counts by 45%.[18] Ferguson et al.[15] showed Fitbit-Zip and Fitbit-One slightly over-estimated step counts by 464 steps/day and 458 steps/day, respectively, when compared with an accelerometer. However, the authors showed Fitbit-Zip and Fitbit-One greatly over-estimated MVPA by 85.7 min/day and 65.9 min/day, respectively.[15]

Irrespective of the similarity or differences in our findings with previous studies, it is important to understand that there is no universally accepted definition of acceptable degree of error for physical activity wearable devices. Some studies recommended that an acceptable measurement error under controlled conditions or for research purposes is within $\pm$ 3%[18,19] and under free-living conditions is within $\pm$ 10%.[18,19] Other studies recommended that mean errors of less than 20% had acceptable validity for clinical purposes.[17] Subsequently, considering the practical significance of the mean difference from the criterion is essential.[18] Our results showed that Fitbit-Flex recorded steps to within 18%, 7% and 13% of the Actigraph steps for males, females and total participants, respectively. Similarly, Fitbit-Flex recorded minutes of MVPA to within 15%, 4% and 10% of the Actigraph minutes of MVPA for males, females and total participants, respectively. Of note, within subgroup analyses, Fitbit-Flex demonstrated greater accuracy for female participants compared with males and total participants. However, the mean absolute per cent errors demonstrated acceptable validity (i.e. $<$ 20%)[17] for the three examined groups, indicating Fitbit-Flex is important for clinical settings in measuring step counts and minutes of MVPA. However, for research purposes, caution should be taken when using Fitbit-Flex as the degree of bias across all three groups was greater than the acceptable validity range (i.e. within 3%).[17,18] Importantly, clinicians should be aware that Bland–Altman plots indicated that Fitbit-Flex validity decreases as the amount of steps taken increases.

Fitbit-Flex measurements for step counts and minutes of MVPA indicated no statistically significant difference for effects of day across the four-day monitoring period. Similarly, Ferguson et al.[15] demonstrated that over two days Fitbit-Zip (intraclass correlation coefficient (ICC) = 0.98) and Fitbit-One (ICC = 0.95) had high reliability coefficients for measuring step counts. However, Fitbit-Zip (ICC = 0.36) and Fitbit-One (ICC = 0.46) had low reliability coefficients for measuring minutes of MVPA.[15] The finding in our study suggested Fitbit-Flex is reliable for measuring step counts and minutes of MVPA in cardiac rehabilitation participants.

Males are commonly identified as being more physically active than females,[27,42] advanced age is a significant independent predictor of less physical activity,[43,44] and increasing BMI is associated with decreasing accuracy of activity monitor output.[44,45] However, the regression analysis results in our study did not show any statistically significant effect from these variables in predicting step counts and minutes of MVPA.

Strengths of our study include the fact that Fitbit-Flex was validated in free-living conditions over four days including weekend days. Furthermore, Fitbit-Flex was validated to measure physical activity guideline recommendations (i.e. step counts and minutes of MVPA). Moreover, our study used a representative sample that included a sufficient number of women to allow for comparison with males, a component that is sometimes missed in cardiovascular research.

Limitations

There are limitations in our study to be acknowledged. First, the results of this study cannot be generalised to all patients with CHD, as eligibility criteria included highly motivated participants who had completed a phase II cardiac rehabilitation programme. Second, fidgeting may be misclassified as valid steps by Fitbit-Flex. Third, Fitbit-Flex is mainly limited to walking activities and is unlikely to provide data related to non-ambulatory activity such as cycling, weight training and swimming.

Conclusion

In summary, Fitbit-Flex is a valid, reliable and inexpensive alternative device for activity monitoring specific to predicted attainment of physical activity guideline recommendations (i.e. step counts and minutes of MVPA). It is also useful for monitoring physical activity in cardiac patients and for comparison among individuals. However, caution must be taken when using Fitbit-Flex for research purposes as it slightly over-estimates step counts and MVPA. Nonetheless, Fitbit-Flex is capable of continuously monitoring free-living conditions and for providing valuable physical activity data for clinicians, individuals and researchers to track physical activity levels. Further research needs to be conducted to investigate the accuracy of activity trackers in cardiac rehabilitation settings. However, challenges remain in keeping up with the rapidly evolving activity trackers in the market.

Acknowledgements

We acknowledge Professor Judy Kay and Mr Lie Ming Tang (Information Technology Department, University of Sydney) for Fitbit data assistance. Also, we acknowledge Dr Lina Engelen (Public Health Department, University of Sydney) for Actigraph data assistance. Lastly, we would like to acknowledge Ms Ann Kirkness, Ms Julia Brooke, Ms Kirsten Jennings and Ms Carly Busseler from the study site at North Sydney Local Health District for recruitment assistance.

Declaration of conflicting interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The authors disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: This study was supported by a Seeding Grant from the University of Sydney, Australia.

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